

根号を含む関数の極限

(例1) 次の極限を求めよ。

point

根号を含む不定形 $\frac{0}{0}$ の場合

有理化

$$\begin{aligned}
 (1) \quad & \lim_{x \rightarrow 2} \frac{\sqrt{x+2}-2}{x-2} \quad \leftarrow \frac{0}{0} \\
 &= \lim_{x \rightarrow 2} \frac{\sqrt{x+2}-2}{x-2} \cdot \frac{\sqrt{x+2}+2}{\sqrt{x+2}+2} \\
 &= \lim_{x \rightarrow 2} \frac{(x+2)-4}{(x-2)(\sqrt{x+2}+2)} \\
 &= \lim_{x \rightarrow 2} \frac{1}{\sqrt{x+2}+2} \\
 &= \frac{1}{4} //
 \end{aligned}$$

$$\begin{aligned}
 (2) \quad & \lim_{x \rightarrow 0} \frac{x}{\sqrt{1+x}-\sqrt{1-x}} \quad \leftarrow \frac{0}{0} \\
 &= \lim_{x \rightarrow 0} \frac{x}{\sqrt{1+x}-\sqrt{1-x}} \cdot \frac{\sqrt{1+x}+\sqrt{1-x}}{\sqrt{1+x}+\sqrt{1-x}} \\
 &= \lim_{x \rightarrow 0} \frac{x(\sqrt{1+x}+\sqrt{1-x})}{(1+x)-(1-x)} \\
 &= \lim_{x \rightarrow 0} \frac{\sqrt{1+x}+\sqrt{1-x}}{2} \\
 &= 1 //
 \end{aligned}$$

$$\begin{aligned}
 (3) \quad & \lim_{x \rightarrow 2} \frac{x-\sqrt{3x-2}}{\sqrt{x+2}-2} \quad \leftarrow \frac{0}{0} \\
 &= \lim_{x \rightarrow 2} \frac{x-\sqrt{3x-2}}{\sqrt{x+2}-2} \cdot \frac{\sqrt{x+2}+2}{\sqrt{x+2}+2} \cdot \frac{x+\sqrt{3x-2}}{x+\sqrt{3x-2}} \\
 &= \lim_{x \rightarrow 2} \frac{x^2-(3x-2)}{(x+2)-4} \cdot \frac{\sqrt{x+2}+2}{x+\sqrt{3x-2}} \\
 &= \lim_{x \rightarrow 2} \frac{(x-1)(x+2)}{(x-2)} \cdot \frac{\sqrt{x+2}+2}{x+\sqrt{3x-2}} \\
 &= 1 //
 \end{aligned}$$

$$\begin{aligned}
 (4) \quad & \lim_{x \rightarrow 1} \frac{x-1}{\sqrt[3]{x}-1} \quad \leftarrow \frac{0}{0} \quad (a-b)(a^2+ab+b^2)=a^3-b^3 \\
 &= \lim_{x \rightarrow 1} \frac{x-1}{\sqrt[3]{x}-1} \cdot \frac{(\sqrt[3]{x})^2+\sqrt[3]{x}+1}{(\sqrt[3]{x})^2+\sqrt[3]{x}+1} \\
 &= \lim_{x \rightarrow 1} \frac{(x-1)(\sqrt[3]{x^2}+\sqrt[3]{x}+1)}{x-1} \\
 &= \lim_{x \rightarrow 1} (\sqrt[3]{x^2}+\sqrt[3]{x}+1) \\
 &= 3 //
 \end{aligned}$$

(例2) 次の極限を求めよ。

$$(1) \quad \lim_{x \rightarrow \infty} (\sqrt{x^2+x}-x) \quad (2) \quad \lim_{x \rightarrow -\infty} (\sqrt{x^2+x}+x)$$

point

根号を含む不定形 $\infty - \infty$ の場合

有理化

* $\sqrt{\quad}$ の $x \rightarrow -\infty$ はコワイ!

$x = -t$ とおきかえ

$$\begin{aligned}
 (1) \quad & \lim_{x \rightarrow \infty} (\sqrt{x^2+x}-x) \quad \leftarrow \infty - \infty \\
 &= \lim_{x \rightarrow \infty} (\sqrt{x^2+x}-x) \cdot \frac{\sqrt{x^2+x}+x}{\sqrt{x^2+x}+x} \\
 &= \lim_{x \rightarrow \infty} \frac{(x^2+x)-x^2}{\sqrt{x^2+x}+x} \\
 &= \lim_{x \rightarrow \infty} \frac{x}{\sqrt{x^2+x}+x} \\
 &= \lim_{x \rightarrow \infty} \frac{1}{\sqrt{1+\frac{1}{x}}+1} \quad \leftarrow \frac{1}{1+0}+1 \\
 &= \frac{1}{2} //
 \end{aligned}$$

$$\begin{aligned}
 (2) \quad & \lim_{x \rightarrow -\infty} (\sqrt{x^2+x}+x) \quad \leftarrow \infty - \infty \\
 &= \lim_{x \rightarrow -\infty} (\sqrt{x^2+x}+x) \cdot \frac{\sqrt{x^2+x}-x}{\sqrt{x^2+x}-x} \\
 &= \lim_{x \rightarrow -\infty} \frac{(x^2+x)-x^2}{\sqrt{x^2+x}-x} \\
 &= \lim_{x \rightarrow -\infty} \frac{x}{\sqrt{x^2+x}-x} \\
 &= \lim_{x \rightarrow -\infty} \frac{1}{\sqrt{1+\frac{1}{x}}-1} \quad \leftarrow \frac{1}{1-0}-1 \\
 &= -\frac{1}{2} //
 \end{aligned}$$

$\sqrt{A^2} = \begin{cases} A & (A \geq 0) \\ -A & (A < 0) \end{cases}$
 ので
 $\sqrt{x^2+x} = \sqrt{x^2(1+\frac{1}{x})}$
 $= \sqrt{x^2} \sqrt{1+\frac{1}{x}}$
 $= -x \sqrt{1+\frac{1}{x}}$

(別解)

$x = -t$ とおくと, $x \rightarrow -\infty$ のとき $t \rightarrow \infty$ であるから

$$\begin{aligned}
 & \lim_{x \rightarrow -\infty} (\sqrt{x^2+x}+x) \\
 &= \lim_{t \rightarrow \infty} (\sqrt{t^2-t}-t) \\
 &= \lim_{t \rightarrow \infty} (\sqrt{t^2-t}-t) \cdot \frac{\sqrt{t^2-t}+t}{\sqrt{t^2-t}+t} \\
 &= \lim_{t \rightarrow \infty} \frac{(t^2-t)-t^2}{\sqrt{t^2-t}+t} \\
 &= \lim_{t \rightarrow \infty} \frac{-t}{\sqrt{t^2-t}+t} \\
 &= \lim_{t \rightarrow \infty} \frac{-1}{\sqrt{1-\frac{1}{t}}+1} \quad \leftarrow \frac{-1}{1+0}+1 \\
 &= -\frac{1}{2} //
 \end{aligned}$$